

Digital Design *TPGA*



Project

(Single Perceptron) AI Accelerator

(Team)

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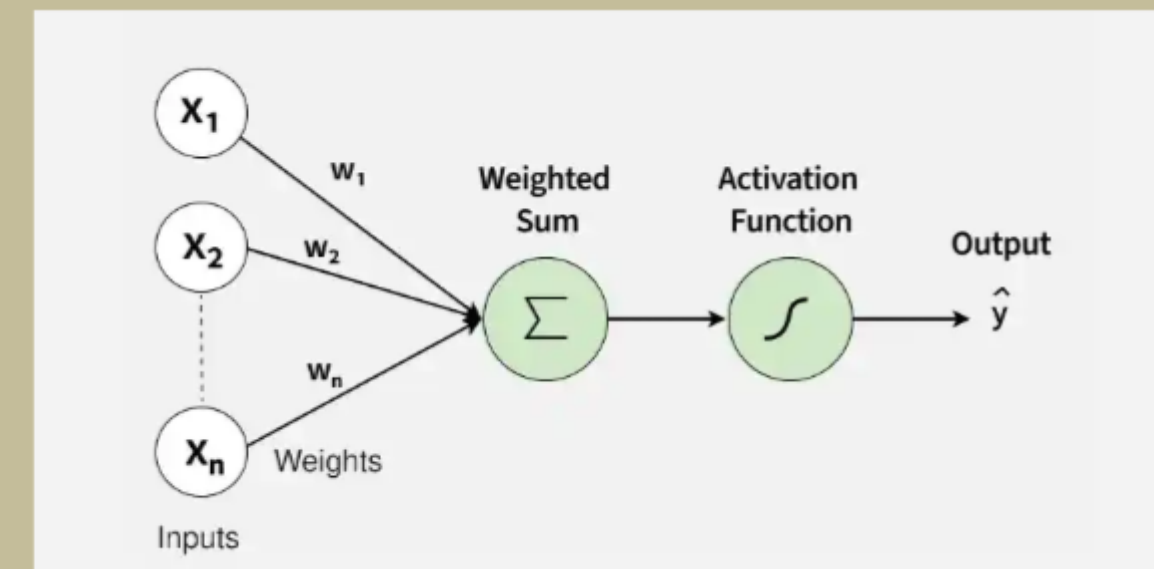
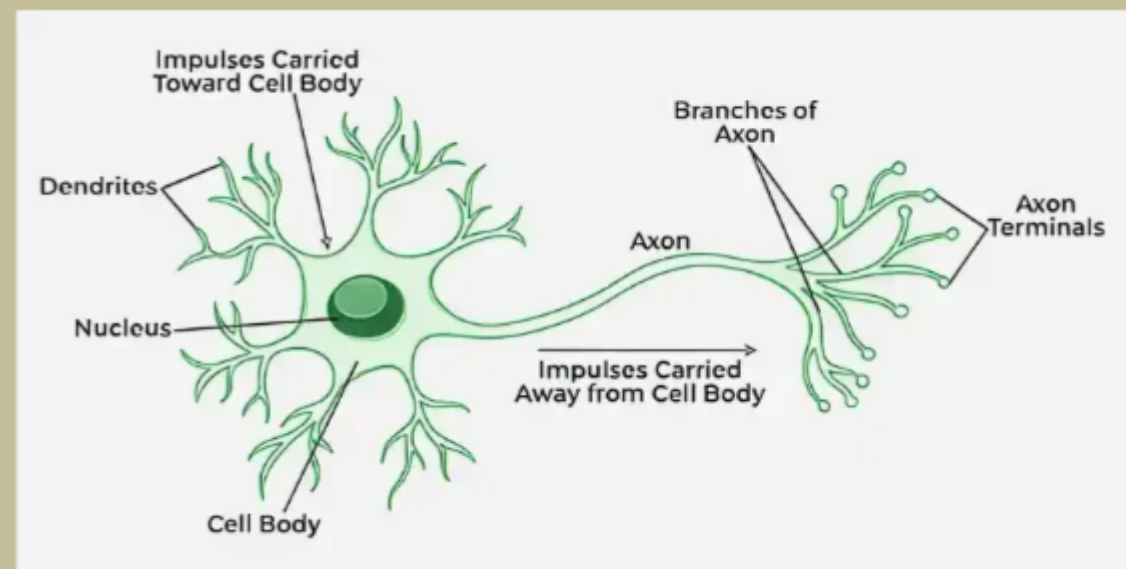
3- Omar Sherif Zaki

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System Concept & Logic

System Concept & Logic



Designing a Single Perceptron Hardware Accelerator.

The core idea:

Biological Mimicry: Translating a neuron into a digital hardware datapath.

MAC Operation: Multiplying inputs by weights and accumulating the sum.

Decision Making: Applying a ReLU activation function to trigger the final output.

```
x_bus ----\
           \
            MUX ---> Multiplier ---> Accumulator ---> Bias Add ---> ReLU ---> Result
           /
w_bus ----/
```

System Concept & Logic

Datapath & MAC Unit

Data Routing (MUX):

Serializes incoming 8-bit features (X_i) and weights (W_i).

Multiplier Stage:

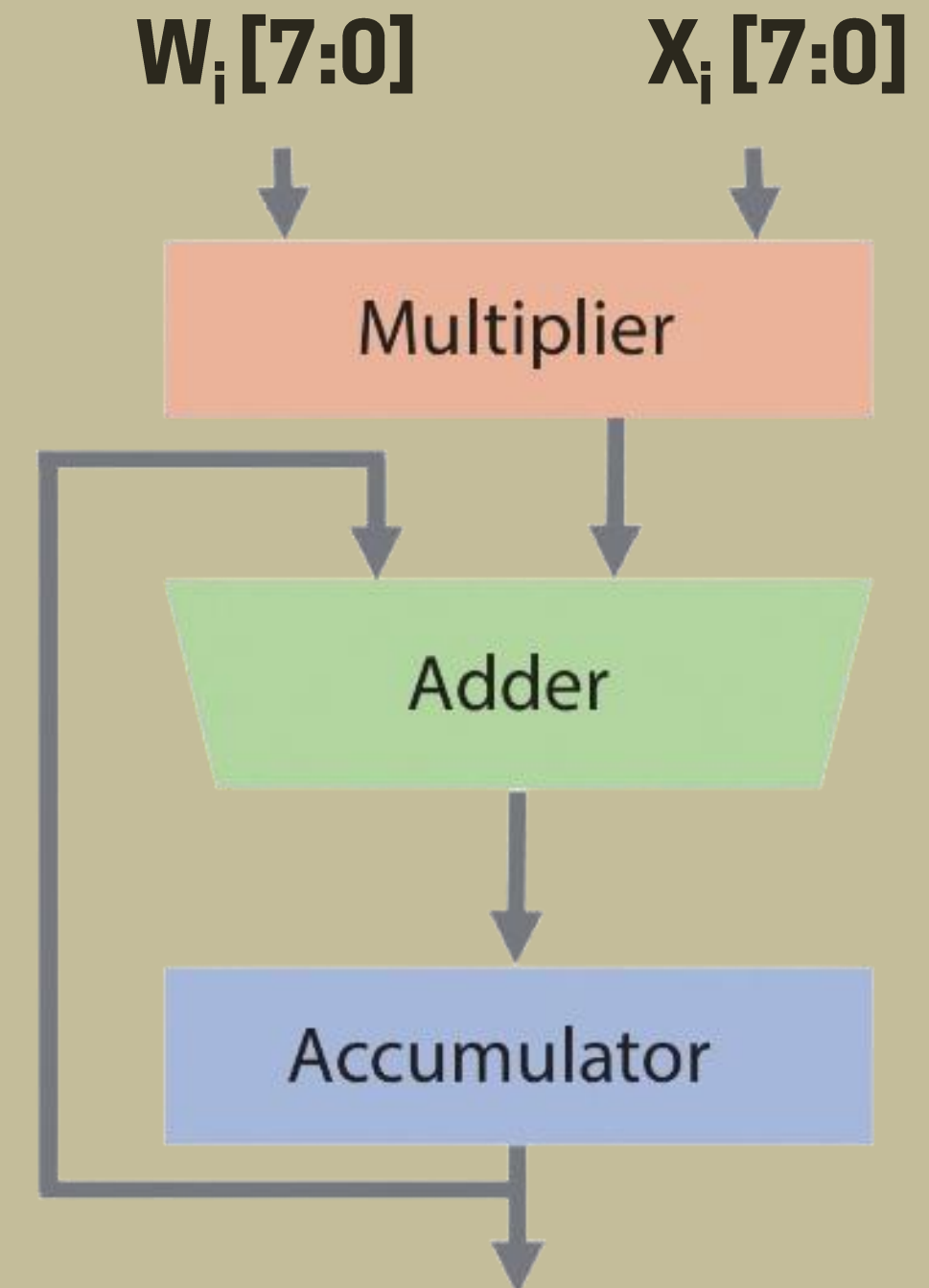
Performs 8x8-bit multiplication, generating a 16-bit product to ensure zero data loss.

Hardware Optimization (Pipelining):

A synchronous register isolates multiplication from accumulation, reducing the critical path and maximizing clock frequency.

The Accumulator:

A 32-bit register designed to safely sum all weighted products without overflow risks.



FSM Control Unit

- State Management:**

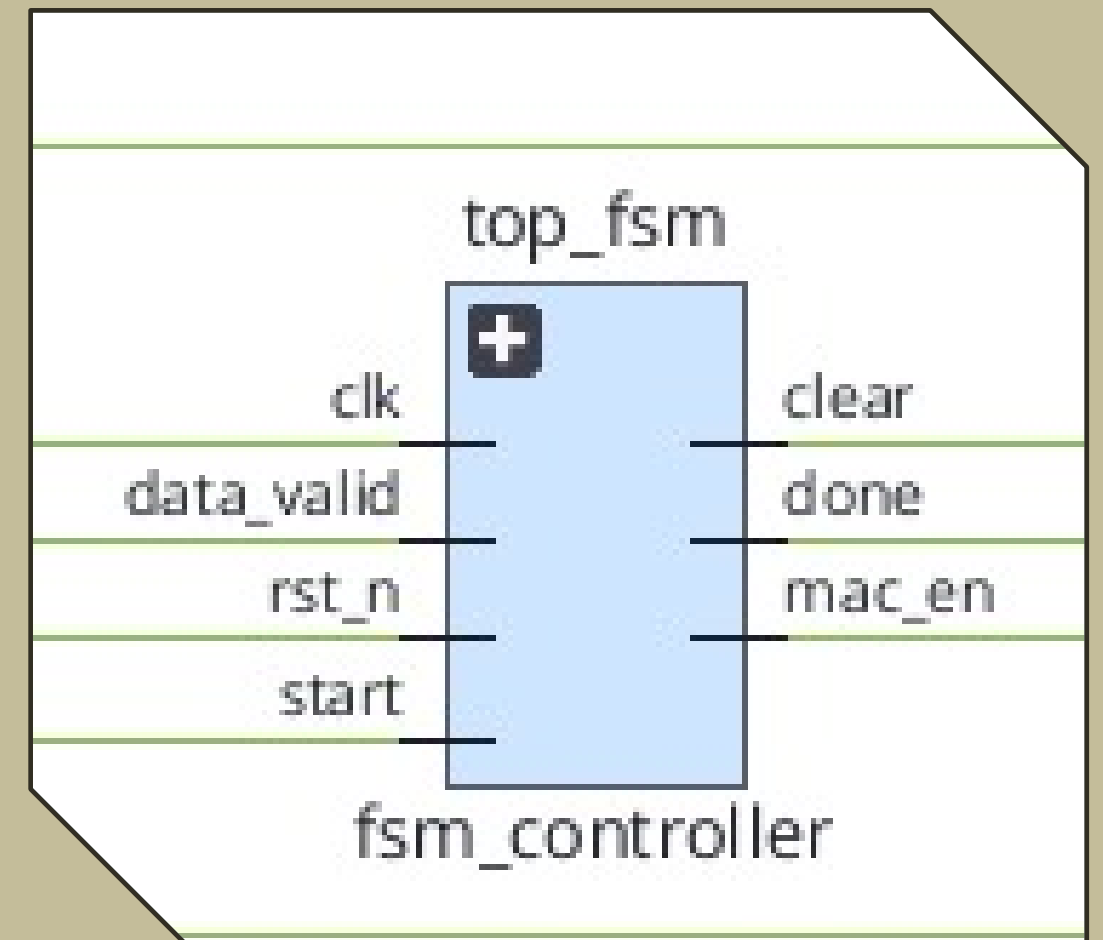
Transitions between IDLE, MAC_LOOP, BIAS_ADD, and DONE.

- Datapath Control:**

Generates enable signals (e.g., mac_en) and drives MUX selectors

- Handshaking:**

Outputs the done flag to signal valid final_output.



Bias & ReLU Activation

Bias Addition:

Adding a constant value to the final accumulated sum to shift the decision boundary.

Hardware-Efficient ReLU:

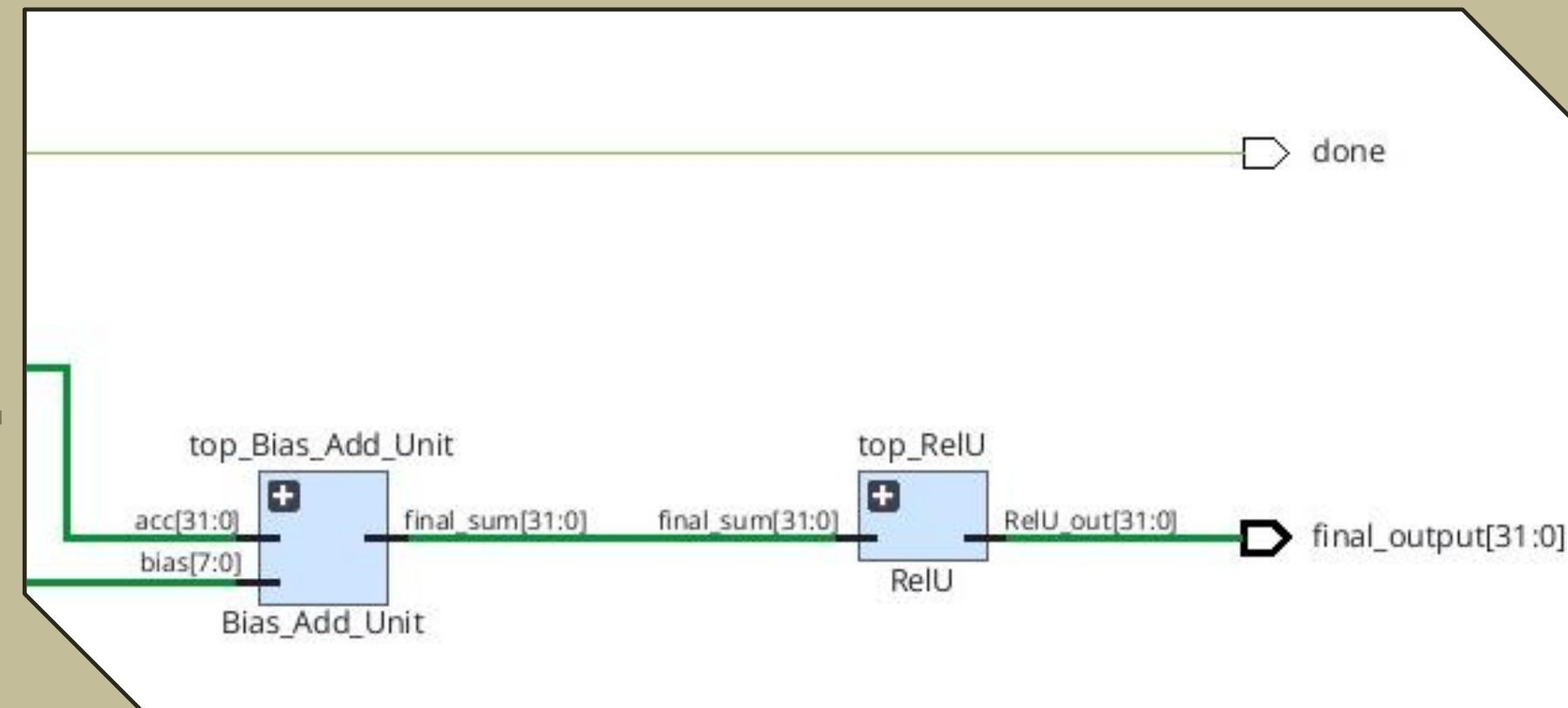
Zero complex mathematical functions used.

Sign-Bit Logic:

Operates simply by checking the Most Significant Bit (MSB) of the signed result:

If MSB == 1 (Negative) → Output = 0

If MSB == 0 (Positive) → Output = Pass unchanged



Top-Level Module

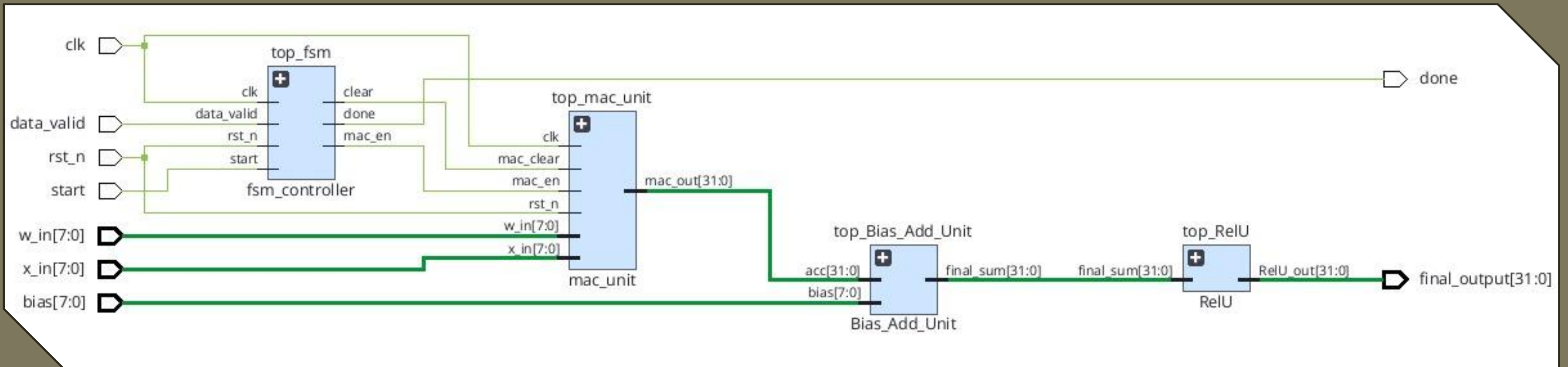
Top-Level Module

System Integration:

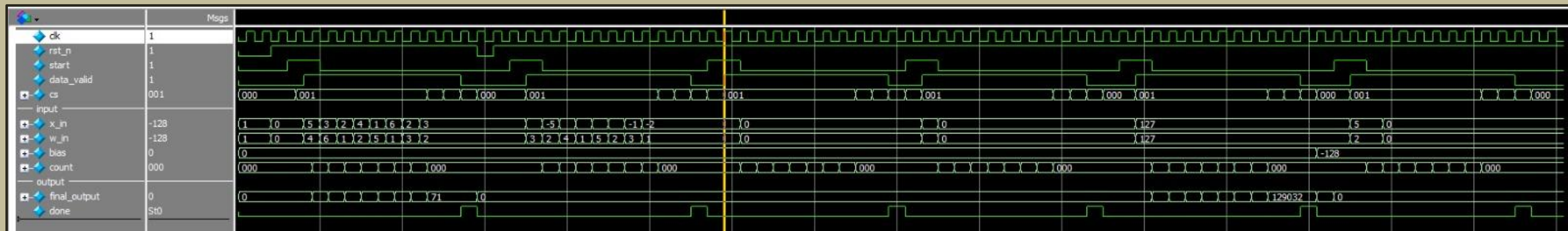
Structural RTLBlock Instantiation: Connecting the Control Unit (FSM) with the Datapath modules (MAC, Bias, ReLU).

Encapsulation:

Hiding internal wiring to provide a clean, unified interface for external systems.



Verification & Simulation Results



Thank you